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OptimWrapper - 5x5 linear and nonlinear systems (CAS-defined)

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OptimWrapper - two 5x5 systems solved from user-defined CAS residuals

GUID 3c9f7e0a-c2b1-4eae-ab68-9ea2c518c510, v1.0.0.0. The plugin has NO built-in test-function ids: the residual systems below are defined here in SMath CAS.

Result layout for every solver: column [x1; ...; xn; f; iterations; ok], hence f = el(res, n+1) = el(res, 6); ok = el(res, n+3) = el(res, 8) (n = 5, ok = 1 means converged).

LINEAR (exact x = [2, -1, 3, 1, -2]): optim_broyden on the 5x1 residual matrix.

NONLINEAR (exact x = [1, 2, -1, 1, 2]): plain Broyden OVERFLOWS here from a non-root start (ResultIsAboveMaxAllowedPositiveNumber), so the sum of squared residuals is minimised with optim_lbfgs (f ~ 0). Solver variables are x1..x5 (not x, the plot axis).

2. Initial guesses for the two systems

```
Linear start x0 = [0, 0, 0, 0, 0] (used by optim_broyden).
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$$s_lin := \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

```
Nonlinear start x0 = [1, 1, 1, 1, 1] (used by optim_lbfgs).
```

$$s_nl := \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

3. Linear 5x5 system - optim_broyden on the residual matrix F(x) = LHS - RHS

Expected root [2, -1, 3, 1, -2]; f = max|ri| ~ 0; ok = 1.

```
Root of the 5x5 linear system from [0,0,0,0,0] -> [2, -1, 3, 1, -2].
```

$$r5lin := \mathrm{optim_broyden}\left(\begin{bmatrix} 2\cdot x1 + 3\cdot x2 - x3 + 2\cdot x4 + x5 + 2 \\ x1 - 2\cdot x2 + 4\cdot x3 - x4 + 3\cdot x5 - 9 \\ 3\cdot x1 + x2 + 2\cdot x3 - 3\cdot x4 + x5 - 6 \\ x1 + 4\cdot x2 - x3 + 2\cdot x4 - x5 + 1 \\ 2\cdot x1 + x2 + 3\cdot x3 + x4 + 2\cdot x5 - 9 \end{bmatrix}, \begin{bmatrix} x1 \\ x2 \\ x3 \\ x4 \\ x5 \end{bmatrix}, s_lin\right) = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 1 \\ -2 \\ 1.9\cdot 10^{-13} \\ 14 \\ 1 \end{bmatrix}$$

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Linear root x1 = 2.

$xL1 := r5lin_1$

Linear root x2 = -1.

$xL2 := r5lin_2$

Linear root x3 = 3.

$xL3 := r5lin_3$

Linear root x4 = 1.

$xL4 := r5lin_4$

Linear root x5 = -2.

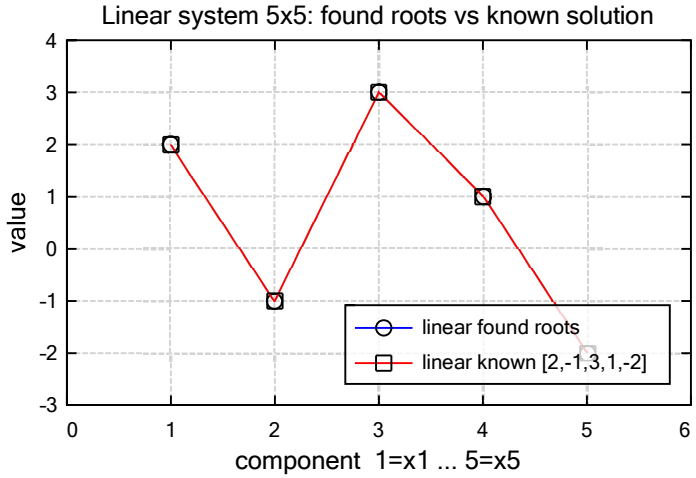
$xL5 := r5lin_5$

Linear residual measure f = max|ri| (expected ~ 0).

$fL := r5lin_6$

Linear convergence flag (expected 1).

$okL := r5lin_8$



$$\begin{cases} \text{augment}\left(\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix}, \begin{bmatrix} r5lin_1\\ r5lin_2\\ r5lin_3\\ r5lin_4\\ r5lin_5 \end{bmatrix}\right) \\ \text{augment}\left(\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix}, \begin{bmatrix} 2\\ 0-1\\ 3\\ 1\\ 0-2 \end{bmatrix}\right) \end{cases}$$

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4. Nonlinear 5x5 system - optim_lbfgs on the sum of squared residuals

Broyden diverges on this system, so SSE = r1^2 + r2^2 + r3^2 + r4^2 + r5^2 is

minimised directly. Expected root [1, 2, -1, 1, 2]; SSE ~ 0; ok = 1.

$$eq := \begin{bmatrix} x1^2 + x2^2 - x3 - 3\cdot x4 - 3 \\ x1\cdot x2 + x3^2 + 2\cdot x5 - 7 \\ x2^2 - x1\cdot x3 - x4 - x5 - 2 \\ x1^3 + x2\cdot x3 - 2\cdot x4 + x5 + 1 \\ x3^2 + x4^2 + x1\cdot x5 - 4 \end{bmatrix}$$

Least-squares root of the 5x5 nonlinear system from [1,1,1,1,1] -> [1, 2, -1, 1, 2].

$$r5nl := \text{optim_lbfgs}\left(\left(eq_1\right)^2 + \left(eq_2\right)^2 + \left(eq_3\right)^2 + \left(eq_4\right)^2 + \left(eq_5\right)^2, \begin{bmatrix} x1 \\ x2 \\ x3 \\ x4 \\ x5 \end{bmatrix}, s_nl\right) = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 1 \\ 2 \\ 1.3077\cdot 10^{-18} \\ 27 \\ 1 \end{bmatrix}$$

Nonlinear root x1 = 1.

$$xN1 := r5nl_1$$

Nonlinear root x2 = 2.

$$xN2 := r5nl_2$$

Nonlinear root x3 = -1.

$$xN3 := r5nl_3$$

Nonlinear root x4 = 1.

$$xN4 := r5nl_4$$

Nonlinear root x5 = 2.

$$xN5 := r5nl_5$$

Nonlinear residual measure f = SSE (expected ~ 0).

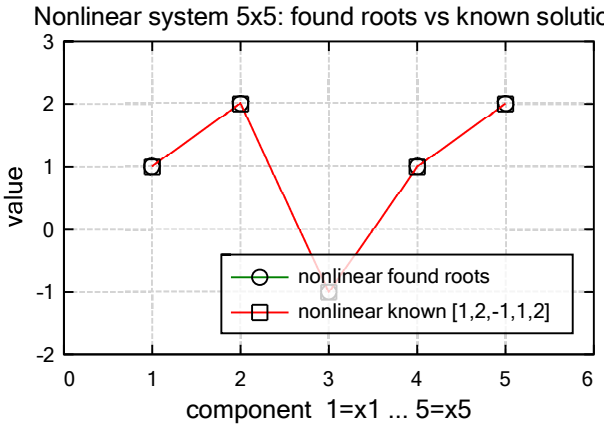
$$fN := r5nl_6$$

Nonlinear convergence flag (expected 1).

$$okN := r5nl_8$$

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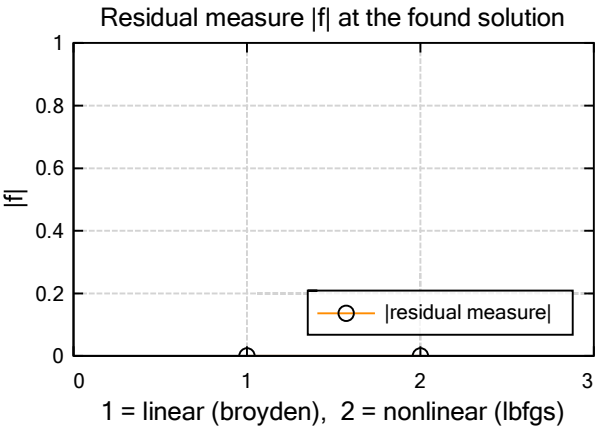


$$\begin{cases} \text{augment}\left(\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix}, \begin{bmatrix}r5nl_1\\r5nl_2\\r5nl_3\\r5nl_4\\r5nl_5\end{bmatrix}\right) \\ \text{augment}\left(\begin{bmatrix}1\\2\\3\\4\\5\end{bmatrix}, \begin{bmatrix}1\\2\\0-1\\1\\2\end{bmatrix}\right) \end{cases}$$

5. Residual measure at the found point (|f|) for the two solvers

Both should be ~ 0: Broyden reports max|ri| for the linear system,

L-BFGS reports the sum of squares for the nonlinear system.



$$\begin{cases} \text{augment}\left(\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}|r5lin_6|\\|r5nl_6|\end{bmatrix}\right) \end{cases}$$

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